

3rd UNDERGRADUATE RESEARCH SYMPOSIUM



Performance and stability of elite wheat genotypes under abiotic stress conditions



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INTRODUCTION



- One of the most important cereal crop in the world
- Third most important cereal of Nepal

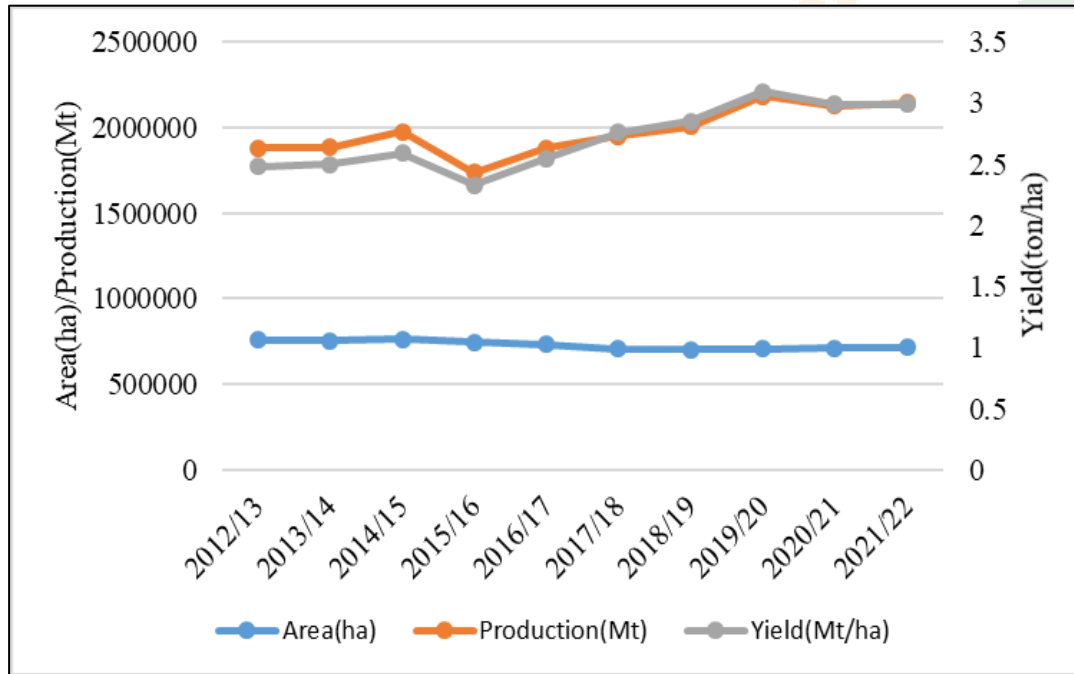


Fig 1. Total production, productivity and area under wheat cultivation in Nepal from 2012/13 to 2021/22 (MOALD, 2022).

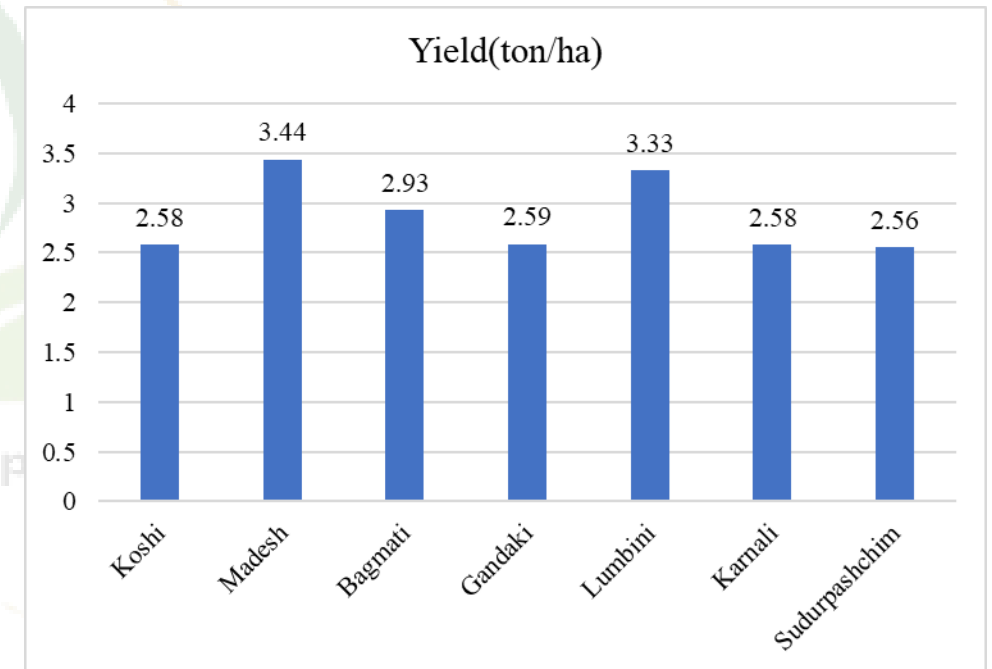


Fig 2. Wheat Productivity of different provinces of Nepal in 2021/22 (MOALD, 2022).



INTRODUCTION

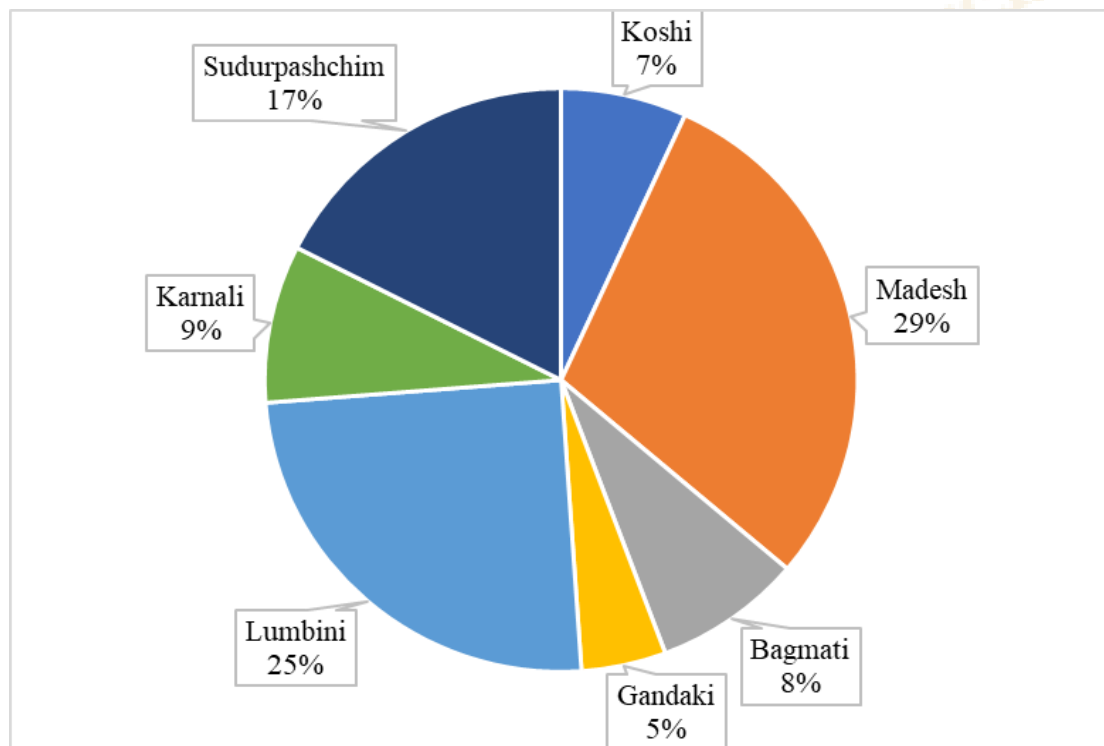


Fig 3. Production share of different provinces in 2021/22 (MOALD, 2022).

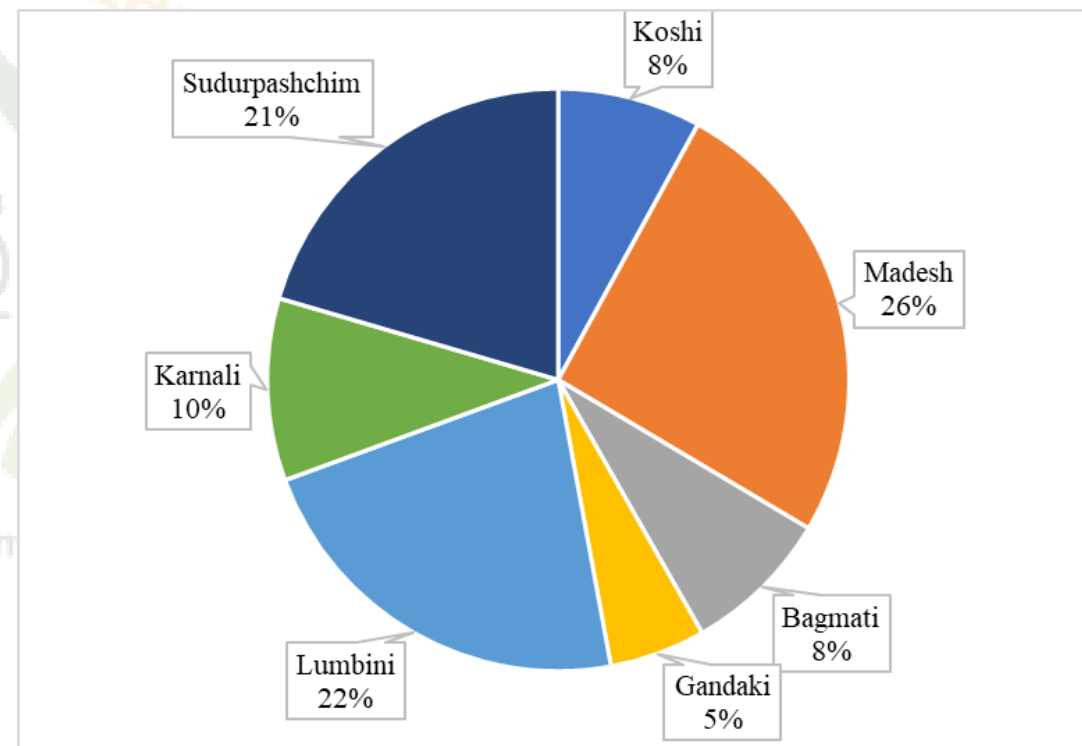


Fig 4. Area share of different provinces in 2021/22 (MOALD, 2022).



STATEMENT OF THE PROBLEM



- Area under Rainfed: 51% (Pandey et al., 2022)
- Area under Heat Stress: 20-27% (Tiwari et al., 2019)
- Yield reduction under
 - Heat stress condition 44-47% (Poudel et al., 2020)
 - Drought 50-60% (Zhao et al., 2020; Poudel et al., 2020)
- Lack of climate resilient wheat genotypes
- On Monetary term annual loss of 1.84 and 1.24 million US \$ with cumulated 3.08 million US \$ due to heat stress and drought
- Temperature rise @ 0.06 °C per year (Paudel et al., 2020)
- Precipitation decline @ 16.09mm (Paudel et al., 2020)

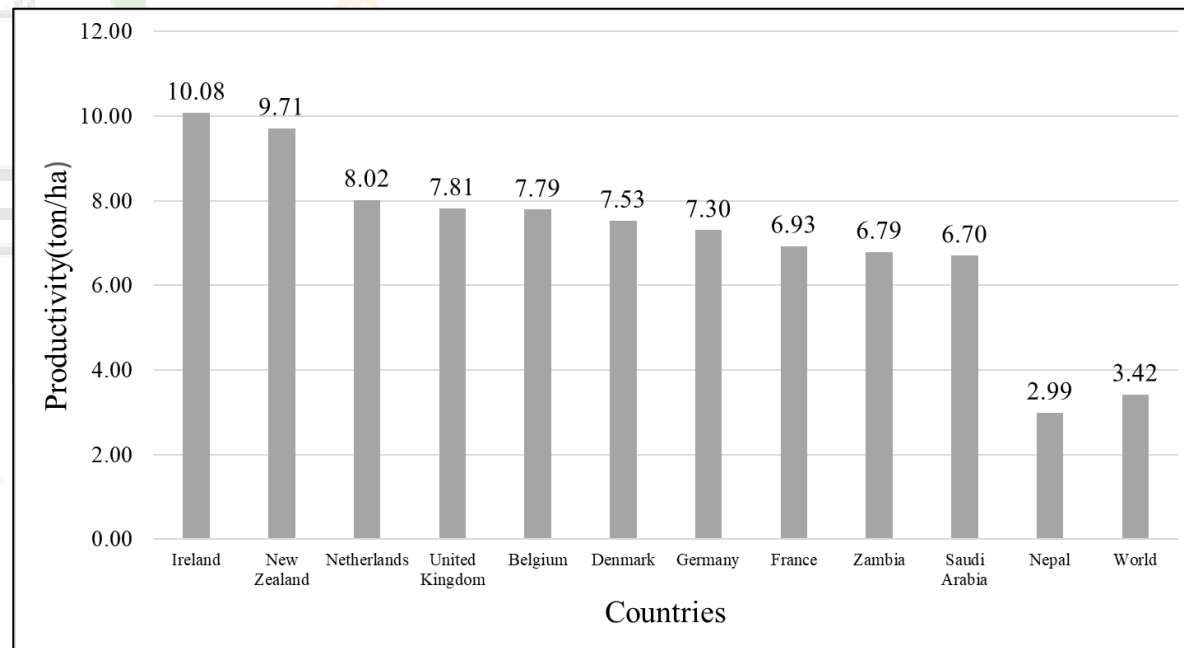


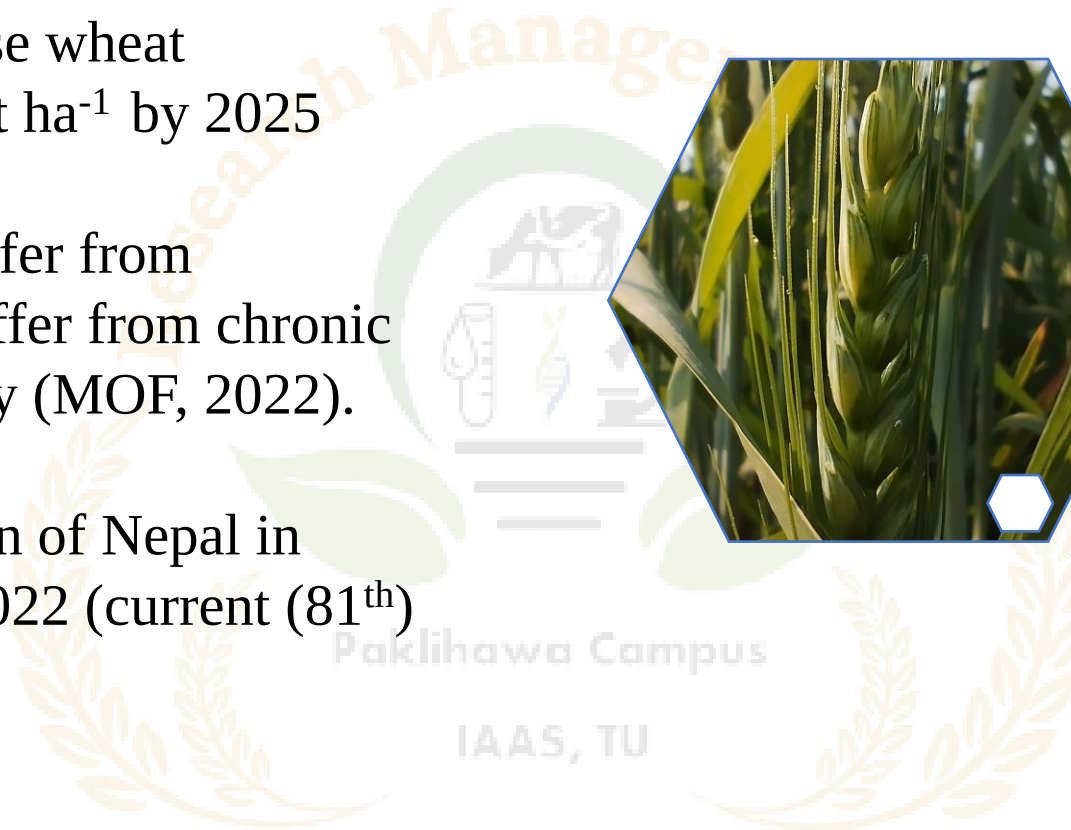
Fig 5. Wheat productivity of different countries (FAOSTAT, 2022).



RATIONALE OF THE STUDY



- Seed Vision aims to raise wheat productivity up to 3.10 t ha^{-1} by 2025
- 16.67% of Nepalese suffer from malnutrition and 6% suffer from chronic micronutrient deficiency (MOF, 2022).
- Improvement of position of Nepal in Global Hunger Index 2022 (current (81th))





OBJECTIVES



Broad objective:

- To determine the performance and stability of wheat genotype under abiotic stress conditions

Specific objectives:

- To evaluate the performance of wheat genotypes under irrigated, rainfed, drought and heat stress conditions
- To identify the yield adaptability and stability of elite wheat genotypes under stress and non stress conditions
- To identify the drought tolerance genotypes based on stress tolerance indices



HYPOTHESIS

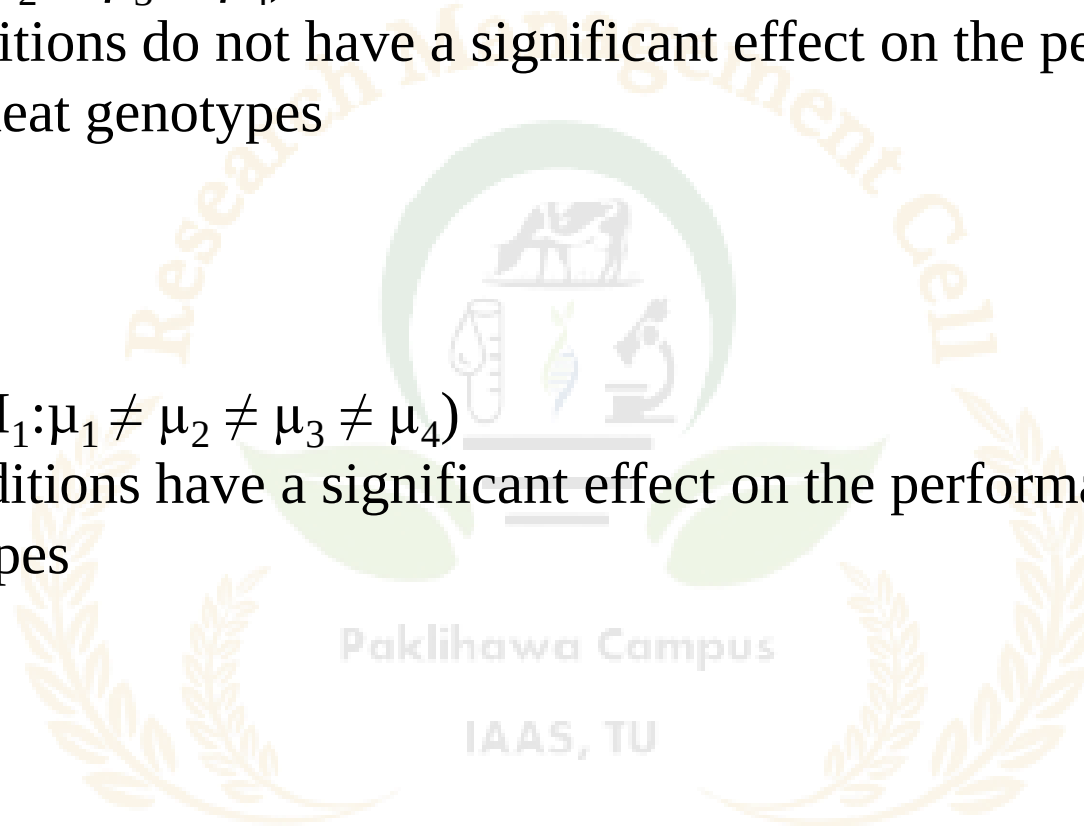


Null hypothesis ($H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$)

- Abiotic stress conditions do not have a significant effect on the performance and stability of elite wheat genotypes

Alternative hypothesis ($H_1: \mu_1 \neq \mu_2 \neq \mu_3 \neq \mu_4$)

- Abiotic stress conditions have a significant effect on the performance and stability of elite wheat genotypes





MATERIALS AND METHODS



- Tropical Terai Region of Nepal
- Latitude: 27°29'02" N
- Altitude: 104 meter above sea level

Longitude: 83°27'17" E

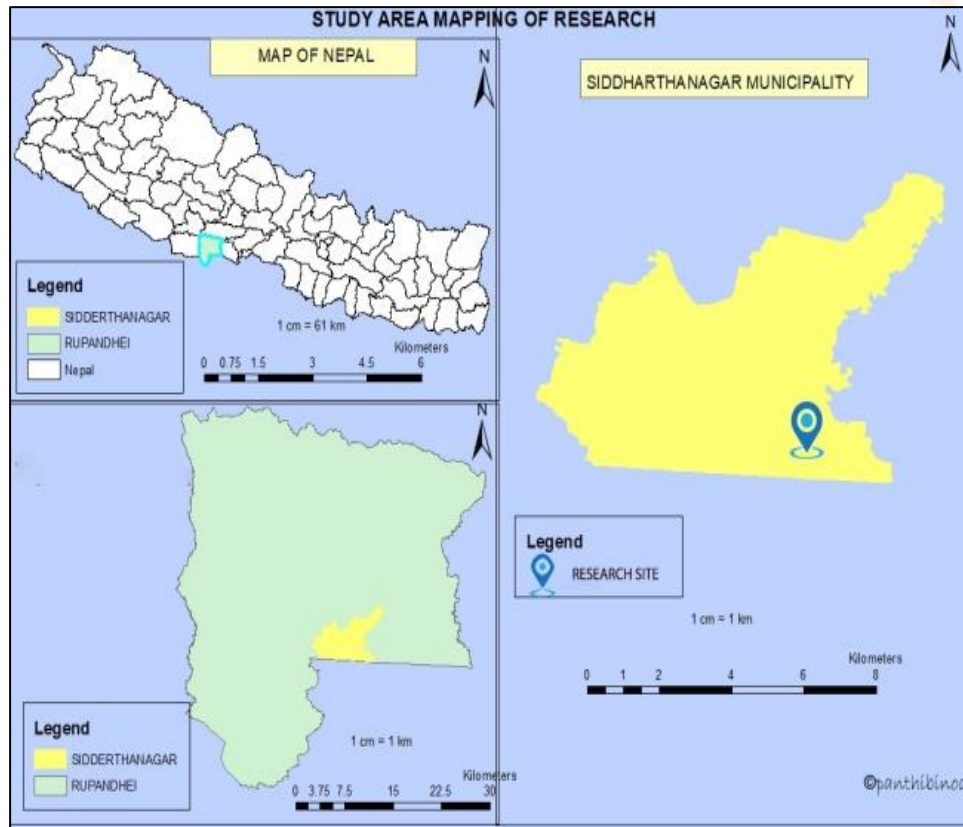


Fig 6. Experimental site

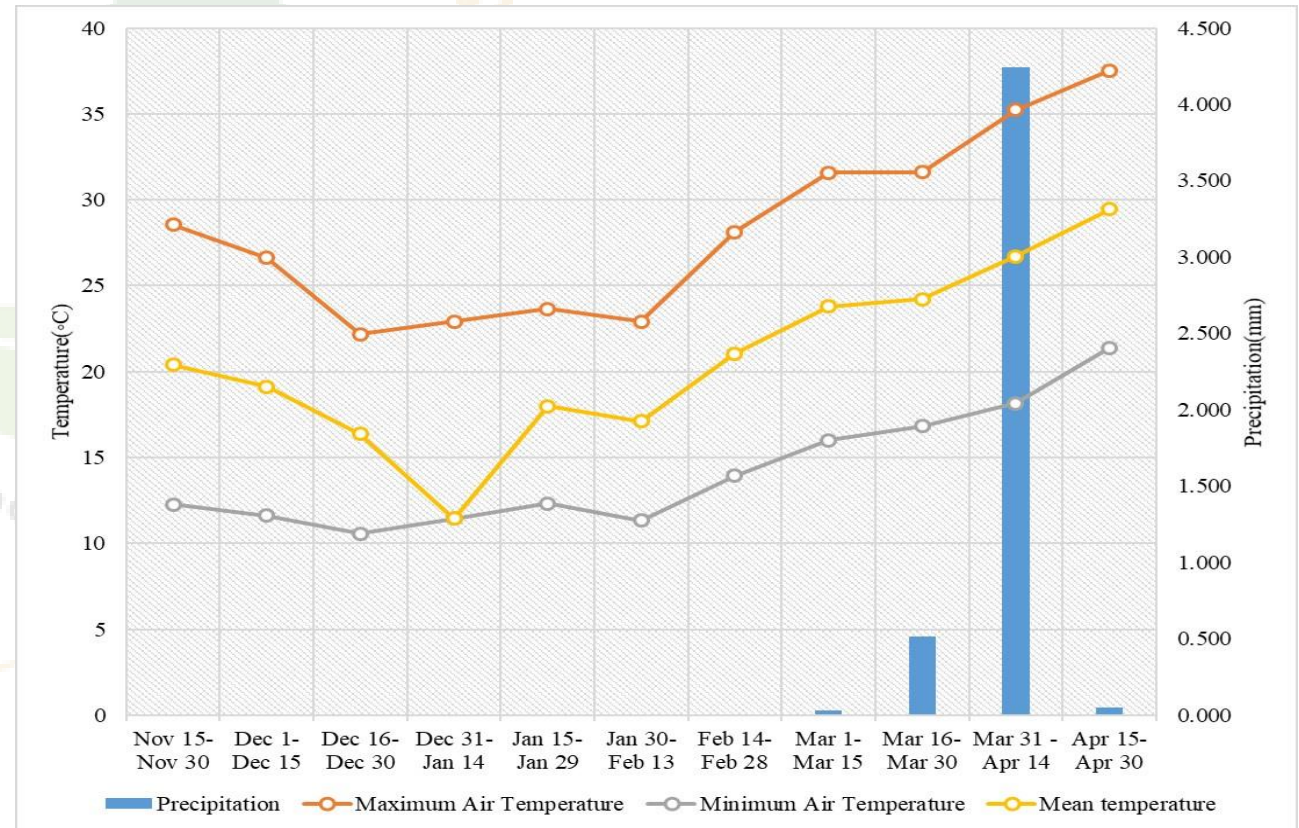


Fig 7. Agrometeorological parameters



PLANT MATERIALIZ



Twenty wheat genotypes comprising three commercial checks viz: Bhrikuti, Gautam and RR 21 were collected from National Wheat Research Program (NWRP).

S.N.	Genotypes	S.N	Genotypes
1	BL 5106	11	NL 1447
2	BL 5099	12	NL 1445
3	BL 4984	13	NL 1506
4	Bhirkuti	14	NL 1504
5	NL 1437	15	NL 1503
6	NL 1402	16	NL 1501
7	Gautam	17	RR 21
8	BL 5116	18	NL 1512
9	NL 1492	19	NL 1509
10	NL 1488	20	NL 1508





LAYOUT OF THE EXPERIMENT



Layout :

- Alpha lattice with two replications
- Twenty experimental units
- Plot size: 2m x 2 m

Replication 1			
T ₁	T ₂	T ₃	T ₄
T ₈	T ₇	T ₆	T ₅
T ₉	T ₁₀	T ₁₁	T ₁₂
T ₁₆	T ₁₅	T ₁₄	T ₁₃
T ₁₇	T ₁₈	T ₁₉	T ₂₀

Replication 2			
T ₁₇	T ₁₈	T ₁₉	T ₂₀
T ₁₆	T ₁₅	T ₁₄	T ₁₃
T ₉	T ₁₀	T ₁₁	T ₁₂
T ₈	T ₇	T ₆	T ₅
T ₁	T ₂	T ₃	T ₄



Fig 8. Experimental Layout



MATERIALS AND METHODS



Four distinct wheat growing conditions

- **Irrigated:** Sown on 30 November, 2022 with recommended six doses of irrigation (NWRP)
- **Rainfed:** Fully depends upon natural precipitation
- **Drought:** Restriction of precipitation by creating a polyhouse
- **Heat Stress:** One month delayed than that of irrigated condition i.e. 30 December, 2022

Data collection

- Days to Booting (DTB)
- Days to Heading (DTH)
- Days to Anthesis (DTA)
- Plant Height (Ph)
- Spike Length (SL)
- Number of Spike Per Meter Square (NSPMS)
- Number of Spikelet Per Spike (NSPS)
- Number Of Grain Per Spike (NGPS)
- Spike Weight (SW)
- Thousand Grain Weight (TGW)
- Grain Yield (GY)





STATISTICAL ANALYSIS



S.N	Software	Analysis
1.	Ms Office (Word, Excel)	Data entry, processing, and stress tolerance indices (STI)
2	SPSS	Combined Analysis of Variance (ANOVA)
3.	GEA-R Version-4 Provided by CIMMYT Mexico	Additive Main Effect and Multiplicative Interaction (AMMI), Genotype, Genotype*Environment Interaction (GGE)
4	Past 4.13	Principal component analysis (PCA)





RESULTS AND DISCUSSION

Table 1. Yield and yield attributing character of wheat genotypes under irrigated condition

Treatment	DTB	DTH	DTA	Ph	SL	NSPMS	NSPS	NGPS	NSW	TGW	GY
BL 5106	75.0	82.0	84.5	96.5	11.1	219.5	18.2	562.5	29.0	39.3	3895
BL 5099	75.5	83.5	84.5	101.7	10.2	291.0	17.0	448.5	29.7	51.0	3975
BL 4984	75.0	83.0	84.0	96.1	9.0	352.0	15.9	417.0	34.8	41.2	4570
Bhrikuti	71.0	75.5	78.5	93.9	9.3	293.5	17.2	433.5	26.6	41.2	3605
NL 1437	72.5	80.0	81.5	93.8	10.1	247.0	16.7	498.5	29.0	47.2	3890
NL 1402	71.5	78.5	80.5	144.6	11.2	294.5	16.8	524.0	33.9	44.1	4460
Gautam	71.0	78.5	80.5	105.5	11.5	262.0	19.1	544.5	31.7	40.3	4200
BL 5116	81.0	84.5	86.5	97.3	10.7	353.0	17.6	558.0	32.4	38.5	4285
NL 1492	73.5	82.0	84.0	100.7	10.7	332.0	17.5	491.5	36.5	40.3	4760
NL 1488	72.0	78.0	81.0	105.6	10.2	260.5	19.1	510.5	39.2	50.5	5075
NL 1447	74.0	80.0	82.5	100.3	10.4	234.5	17.9	551.0	33.6	43.7	4430
NL 1445	80.5	86.0	88.0	101.2	10.0	309.5	17.6	517.5	34.6	45.3	4545
NL 1506	74.0	81.5	82.5	93.9	10.8	289.5	19.7	510.0	36.7	37.8	4785
NL 1504	79.5	85.0	88.0	103.5	11.2	267.5	18.6	514.5	25.9	42.4	3525
NL 1503	74.0	82.5	86.0	95.0	9.1	295.0	17.3	550.5	35.7	39.7	4675
NL 1501	69.0	78.0	79.5	99.3	9.1	323.5	16.8	412.0	33.6	44.1	4430
RR 21	69.0	76.5	80.0	107.0	9.9	310.3	17.4	366.5	26.6	45.6	3610
NL 1512	73.5	78.5	82.5	100.6	9.5	333.0	17.9	496.0	34.0	44.3	4470
NL 1509	73.0	78.0	81.0	94.9	9.2	240.5	19.5	448.0	32.0	51.9	4235
NL 1508	76.0	83.0	85.0	91.5	10.8	257.5	21.1	423.0	26.8	44.2	3630
Grand Mean	74.0	80.7	83.0	101.1	10.2	288.3	17.9	488.9	32.1	43.6	4253
STD	3.34	2.98	2.78	11.15	0.80	39.00	1.26	56.89	3.85	4.11	448.7
CV	4.51	3.69	3.35	11.03	7.84	13.53	7.02	11.64	11.98	9.41	10.55
F-value	***	***	***	ns	ns	ns	*	*	ns	ns	ns



RESULTS AND DISCUSSION

Table 2. Yield and yield attributing character of wheat genotypes under rainfed condition

Treatment	DTB	DTH	DTA	Ph	SL	NSPMS	NSPS	NGPS	NSW	TGW	GY
BL 5106	72.0	80.0	82.5	70.9	8.9	162.5	14.9	324.0	16.6	34.7	1055
BL 5099	72.5	78.0	80.0	77.4	8.1	195.5	13.9	235.0	16.2	40.5	1275
BL 4984	73.5	79.0	80.0	73.3	9.1	203.0	15.6	339.5	16.9	38.1	1500
Bhrikuti	66.5	73.0	75.5	68.8	8.7	222.5	15.3	248.0	18.5	35.5	1610
NL 1437	68.0	77.5	78.0	72.1	8.5	200.5	13.9	314.5	18.3	37.5	1365
NL 1402	67.5	75.5	78.0	78.8	9.2	189.5	15.3	340.5	20.0	40.1	1660
Gautam	67.5	74.5	77.0	77.7	9.9	204.5	13.5	237.0	19.4	40.5	1260
BL 5116	75.0	80.0	82.5	75.5	9.3	205.0	14.5	299.0	17.6	39.3	1430
NL 1492	70.5	78.0	79.0	74.4	8.4	254.8	13.4	308.0	17.3	38.4	1560
NL 1488	66.0	73.0	78.0	78.4	8.8	155.5	15.0	348.5	21.0	42.7	1370
NL 1447	68.0	76.0	78.0	75.2	9.2	148.5	15.8	310.0	23.3	40.3	1340
NL 1445	74.0	81.0	82.0	79.1	8.4	181.0	14.9	369.0	23.8	43.0	1665
NL 1506	70.5	78.0	79.0	76.5	10.4	192.5	20.0	435.0	28.8	43.0	2105
NL 1504	72.5	80.5	83.0	76.1	8.8	223.5	14.7	309.5	19.4	42.6	1830
NL 1503	71.0	78.0	79.0	71.8	7.5	214.0	13.0	303.0	14.5	35.1	1440
NL 1501	67.0	72.0	76.0	71.6	8.1	219.0	12.8	289.5	16.6	39.4	1275
RR 21	65.5	73.0	77.0	76.4	9.5	136.5	14.5	237.5	14.8	40.6	920
NL 1512	69.0	74.5	78.5	77.7	8.3	207.5	13.4	322.5	18.7	38.9	1480
NL 1509	65.5	73.5	76.0	74.7	8.7	134.0	15.3	303.5	20.0	42.6	1430
NL 1508	73.5	79.0	81.0	73.2	8.9	172.0	16.1	321.0	22.4	40.7	2095
Grand Mean	69.8	76.7	79.0	75.0	8.8	191.1	14.8	309.7	19.2	39.7	1483
STD	3.1	2.9	2.3	2.9	0.7	31.5	1.6	47.9	3.4	2.6	296
CV	4.4	3.8	2.9	3.9	7.4	16.5	10.5	15.5	17.8	6.5	19.9
F-Value	***	***	***	ns	ns	ns	*	*	ns	ns	ns



RESULTS AND DISCUSSION

Table 3. Yield and yield attributing character of wheat genotypes under drought condition

Treatment	DTB	DTH	DTA	Ph	SL	NSPMS	NSPS	NGPS	NSW	TGW	GY
BL 5106	74.0	80.5	82.0	73.4	8.7	105.8	15.1	243.0	14.7	23.8	620
BL 5099	74.0	78.5	78.5	72.8	6.7	174.5	11.8	151.0	8.7	30.7	645
BL 4984	70.0	78.5	78.5	66.6	7.0	229.5	10.6	183.0	7.4	22.3	480
Bhrikuti	67.0	73.0	76.5	65.0	7.0	196.8	10.7	237.0	9.4	33.2	765
NL 1437	67.5	76.5	77.0	64.1	7.0	183.0	11.5	202.5	9.2	24.7	495
NL 1402	67.5	75.0	77.0	66.1	6.9	183.0	12.4	194.0	9.3	24.0	620
Gautam	67.0	73.0	75.5	72.6	8.0	178.3	10.3	194.5	9.1	31.6	670
BL 5116	75.0	82.0	83.0	69.6	8.6	163.3	12.8	256.5	11.3	26.8	755
NL 1492	70.5	78.0	78.5	71.6	8.9	187.0	13.1	310.5	13.7	24.0	650
NL 1488	65.5	73.0	75.5	78.9	8.3	143.8	13.9	263.5	13.8	28.4	660
NL 1447	66.5	72.5	75.5	75.5	9.0	182.3	13.7	253.0	12.3	24.7	740
NL 1445	71.5	79.0	79.5	73.8	8.5	125.3	13.6	292.5	13.5	29.6	660
NL 1506	70.5	77.5	78.5	65.9	7.4	159.3	12.6	199.5	10.2	25.5	470
NL 1504	72.5	80.0	81.5	71.0	7.8	127.0	13.3	236.0	9.8	24.9	405
NL 1503	72.5	79.5	81.0	61.7	6.5	181.3	10.3	170.0	6.4	19.2	320
NL 1501	67.0	74.5	76.0	61.3	7.4	153.8	11.1	178.0	8.0	25.4	480
RR 21	65.5	73.0	75.0	75.7	8.9	116.3	12.3	200.0	12.8	29.8	365
NL 1512	69.5	78.0	78.5	73.7	7.1	140.3	12.0	240.5	11.5	30.0	640
NL 1509	67.0	74.0	77.0	68.3	7.7	79.8	12.6	283.5	12.2	28.6	375
NL 1508	71.0	78.5	79.0	64.5	7.8	142.3	13.0	248.5	12.7	27.1	680
Grand Mean	69.6	76.7	78.2	69.6	7.8	157.6	12.3	226.9	10.8	26.7	575
STD	3.0	3.0	2.3	5.0	0.8	35.5	1.3	43.5	2.4	3.5	138
CV	4.3	3.9	3.0	7.2	10.4	22.5	10.6	19.2	22.0	13.1	24.0
F-value	***	***	**	ns	*	***	ns	ns	ns	ns	ns



RESULTS AND DISCUSSION

Table 4. Yield and yield attributing character of wheat genotypes under heat stress condition

Treatment	DTB	DTH	DTA	Ph	SL	NSPMS	NSPS	NGPS	NSW	TGW	GY
BL 5106	71.0	73.5	75.0	66.4	8.8	127.0	15.5	426.5	19.0	32.2	1720
BL 5099	71.0	73.0	74.0	71.7	8.2	225.0	13.8	381.0	16.7	34.1	1825
BL 4984	66.0	71.0	74.0	68.8	8.1	256.0	13.5	288.0	17.2	35.2	2025
Bhrikuti	64.5	70.5	72.5	63.7	9.4	187.0	14.5	239.0	17.6	27.7	2090
NL 1437	64.5	68.5	72.5	64.9	8.0	284.0	13.8	383.0	20.3	35.6	2380
NL 1402	61.0	67.0	71.0	76.5	9.0	232.0	15.5	363.0	20.4	35.2	2805
Gautam	62.0	67.0	71.0	77.7	10.6	204.0	16.1	312.0	19.9	36.1	2300
BL 5116	71.0	74.0	76.0	72.4	9.6	214.0	15.8	368.5	21.0	34.8	2155
NL 1492	65.0	69.5	72.5	83.2	8.8	234.5	14.8	411.5	23.8	28.9	2035
NL 1488	60.0	66.5	71.0	73.0	8.9	209.5	15.0	383.5	25.5	40.8	2920
NL 1447	61.0	66.5	71.0	73.7	8.9	237.5	15.1	367.0	22.3	33.8	2325
NL 1445	69.0	72.0	74.0	74.5	8.4	251.0	14.7	431.0	21.0	26.0	2555
NL 1506	66.5	72.0	74.0	80.7	9.3	206.5	16.5	431.5	25.4	34.0	2715
NL 1504	71.0	74.0	75.0	80.4	8.5	233.0	15.9	345.0	19.4	36.6	2435
NL 1503	64.5	72.5	72.5	81.0	9.4	238.0	15.0	367.0	21.2	32.8	2600
NL 1501	60.5	66.5	71.0	82.6	10.2	247.0	14.6	410.0	21.7	34.5	2525
RR 21	61.5	68.0	71.0	85.4	8.7	240.5	14.8	346.5	17.5	29.6	1975
NL 1512	62.0	70.0	72.5	86.2	8.8	228.5	13.9	387.5	19.8	33.8	2845
NL 1509	62.0	69.5	72.5	81.4	9.2	195.0	14.9	350.5	21.6	38.4	2165
NL 1508	69.0	72.5	74.5	77.1	8.9	218.0	14.8	375.0	22.5	34.5	2020
Grand Mean	65.2	70.2	72.9	76.0	9.0	223.4	14.9	368.4	20.7	33.7	2321
STD	3.9	2.7	1.6	6.7	0.7	32.0	0.8	47.9	2.5	3.5	348
CV	6.1	3.8	2.2	8.8	7.3	14.3	5.4	13.0	12.0	10.4	15.0
Significance	***	***	**	***	***						



RESULTS AND DISCUSSION

Table 5: Combined analysis of variance of yield and yield attributes under tested wheat growing environment of Bhairahawa, Rupandehi, Nepal

	DTB	DTH	DTA	Ph (cm)	SL (cm)	NSPMS (frequency)	NSPS (frequency)	NGPS (frequency)	NSW (g)	TGW (g)	GY (kg ha ⁻¹)
Irrigated	74.0±3.49 ^a	80.7±3.15 ^a	83.0±2.97 ^a	101.1±16.63 ^a	10.2±0.99 ^a	288.3±50.27 ^a	17.9±1.44 ^a	488.9±67.26 ^a	32.1±5.59 ^a	43.6±5.13 ^a	4253±652.3 ^a
Rainfed	69.8±3.13 ^b	76.7±3.11 ^b	79.0±2.35 ^b	75.0±4.05 ^b	8.8±0.82 ^b	191.1±38.76 ^c	14.8±1.65 ^b	309.7±62.39 ^c	19.2±4.13 ^b	39.7±3.07 ^b	1483±355.5 ^c
Drought	69.6±3.26 ^b	76.7±3.20 ^b	78.2±2.64 ^b	69.6±7.57 ^c	7.8±0.94 ^c	157.6±37.75 ^d	12.3±1.58 ^c	226.9±59.39 ^b	10.8±3.01 ^c	26.9±4.97 ^d	575±191.3 ^d
Heat Stress	65.2±4.23 ^c	70.2±2.83 ^c	72.9±1.80 ^c	76.0±7.10 ^b	9.0±0.70 ^b	223.4±38.72 ^b	14.9±1.61 ^d	368.4±59.09 ^d	20.7±3.22 ^b	33.7±4.27 ^c	2321±609.3 ^c
Grand Mean	69.6±4.72	76.1±4.86	78.3±4.38	80.4±15.75	8.9±1.22	215.1±63.62	15.0±2.53	348.5±113.77	20.7±8.65	36.1±7.68	2158±1446.0
Coefficient of variation	5.20	5.72	5.33	17.50	11.16	25.89	15.28	31.61	42.41	20.26	72.66379
F- value	***	***	***	***	***	***	***	***	***	***	***



RESULTS AND DISCUSSION

Table 6: Percentage loss of yield and yield attributes under tested wheat growing conditions as compared to irrigated.

% Reduction	DTB	DTH	DTA	Ph (cm)	SL (cm)	NSPMS (frequency)	NSPS (frequency)	NGPS (frequency)	NSW (g)	TGW (g)	GY (kg ha ⁻¹)
in Rainfed	5.7	5.0	4.8	25.9	13.4	33.7	17.5	36.6	40.3	9.1	65.1
in drought	6.0	5.0	5.8	31.2	23.9	45.3	31.2	53.6	66.4	38.4	86.5
in heat stress	12.0	13.0	12.2	24.8	11.9	22.5	16.7	24.7	35.5	22.7	45.4

S.N	Environment	Yield Reduction	Reason
1	Rainfed	65%	Low soil moisture inhibits nutrient uptake and affects photosynthate assimilation of wheat (Gajghate et al., 2020).
2	Drought	86%	Soil water deficit results in poor germination and causes tiller mortality (Kajla, 2015).
3	Heat Stress	45%	Formation of reactive oxygen species (ROS) (Foyer, 2018). High temperature during reproductive stage causes pollen abortion resulting poor grain formation (Qaseem et al., 2019)



AMMI Model ANOVA



Table 8. Additive main effect and multiplicative interaction (AMMI) model analysis of variance

	df	SS	MS	F	p value	% Explained
ENV	3	295.02	98.34	372.49***	0	94.76
GEN	19	6.62	0.35	1.32	0.19	2.13
ENV*GEN	57	9.68	0.17	0.64	0.95	3.11
PC1	21	5.70	0.27	1.44	0.15	58.81
PC2	19	2.40	0.13	0.67	0.82	24.81
Residuals	80	21.12	0.264	NA	NA	0

*** denotes significant at 0.01% ($p < 0.001$); ENV= Environment; GEN= Genotype; ENV*GEN= Environment*Genotype; PC1= Principal Component 1; PC2= Principal Component 2; df= Degree of freedom; SS= Sum of square; MS= Mean square;

Similar effect of environment on grain yield was observed by Nagailo et al., 2019 and Mohammadi et al., 2018.



AMMI Biplot



- AMMI model was also used by El-salam & Asran, 2018.
- The genotypes close to the origin: stable genotypes (El-salam & Asran, 2018).
- Genotype lies close to arrowhead were adaptable genotypes
- **Stable genotype:** 8 (BL 5116), 7 (Gautam), 19 (NL 1509)
- **Adaptable genotype:**
 - ❖ Irrigated conditions(A): 10 (NL1488)
 - ❖ Heat stress conditions(B):18 (NL 1512)
 - ❖ Rainfed conditions(C) : 4 (Bhrikuti)
 - ❖ Drought conditions(D) : 17 (RR 21)

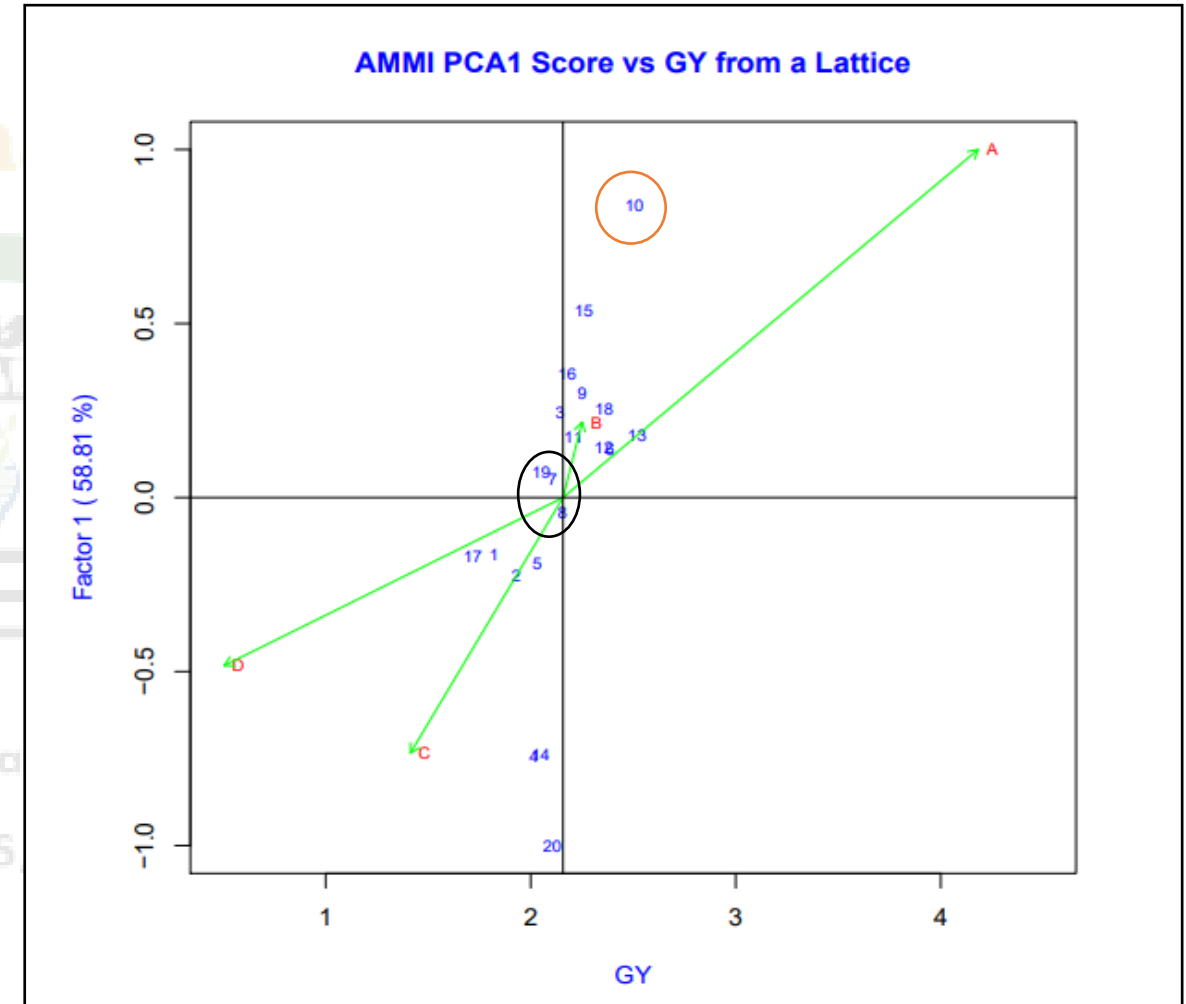


Fig 9. AMMI PCA1 Score vs GY from a Lattice



Which-Won-Where Model



- Which-Won-Where (WWW) model was used by Mungal et al., 2020.
- Genotypes lies on the center of the biplot was most stable across all tested conditions (Gupta & Kumar, 2019).
- Vertex genotypes were the most stable genotype (Poudel et al., 2020).
- **Stable genotype:** 19 (NL 1509), 12(NL1445),17(RR21), and 7(Gautam).
- **Adaptable genotype:**
 - Irrigated conditions:10 (NL1488)
 - Heat stress conditions : 6 (NL 1402),13 (NL 1506), 18 (NL 1512)
 - Rainfed conditions : 20 (NL 1508)
 - Drought conditions : 20 (NL 1508)

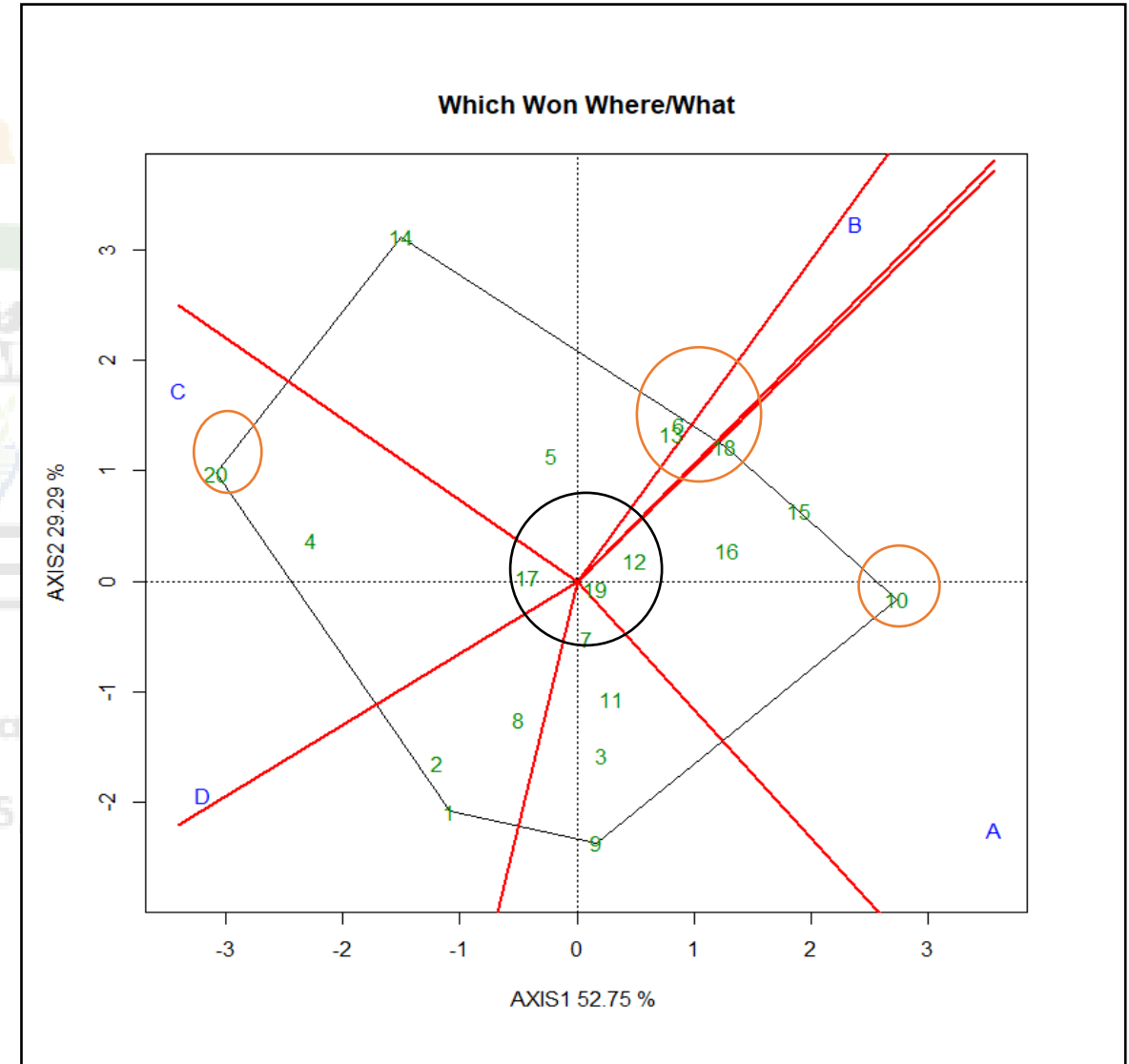


Fig 10. WWW model



RESULTS AND DISCUSSION

- Mean vs. Stability model was also used by Apraku et al., 2016 and Aktas, 2016.
- High yielding: Lies towards the side of arrowhead (Poudel, 2019).
- Stability: The length of projection on the AEC (Average Environment Coordinates) abscissa (Jat et al., 2017).
- **High yielding and stable:** 17 (RR 21), 5 (NL 1437), 14 (NL 1504), 13 (NL 1506).

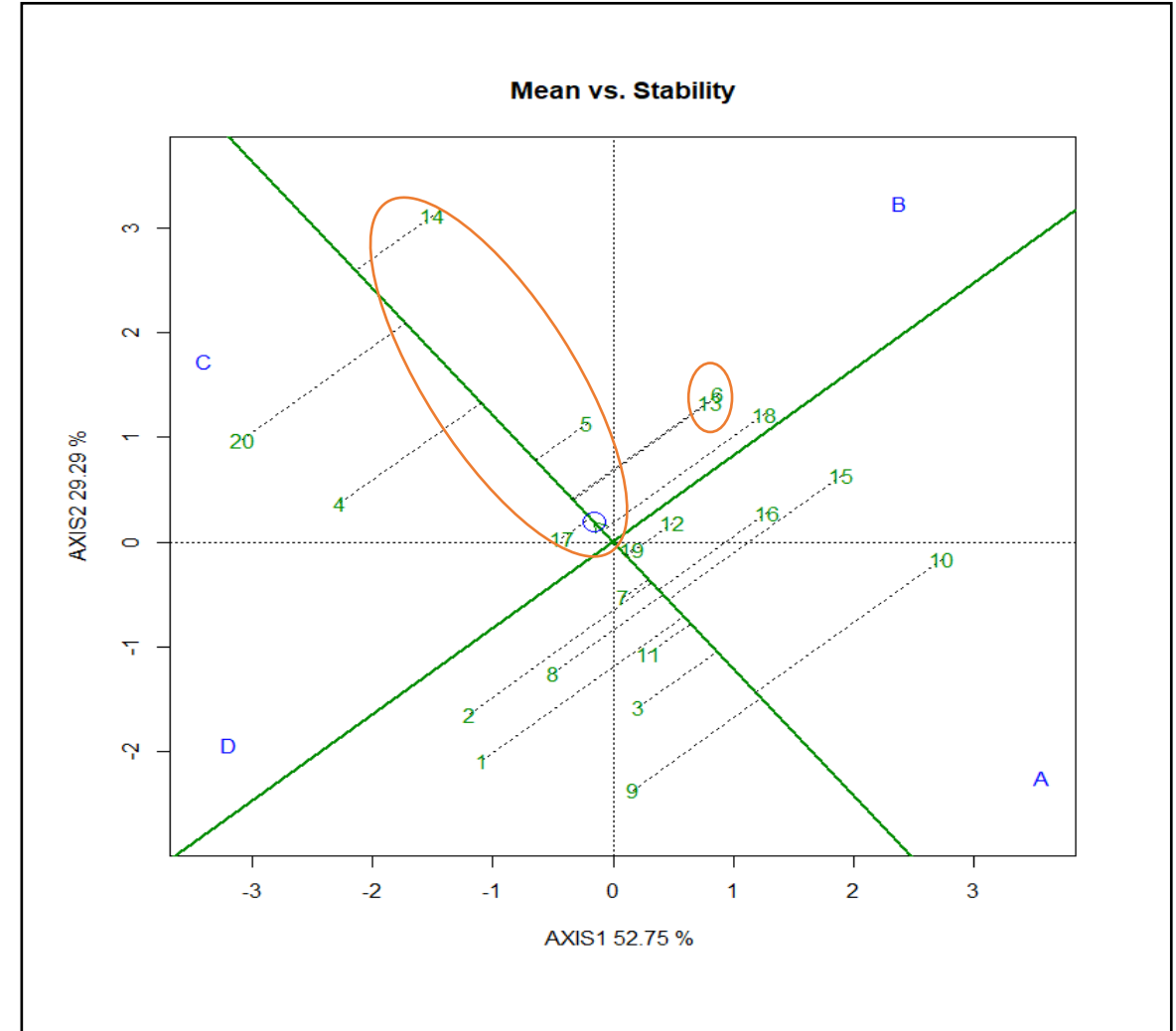


Fig 11. Mean vs stability model



STRESS TOLERANCE INDICES (STIs)

Table 7. Stress tolerance indices of wheat genotypes under drought conditions.

Genotypes	Y _p	Y _s	TOL	YSI	MP	GMP	SSI	STI
BL 5106	3.90	0.62	3.28	0.16	2.26	1.55	0.97	0.13
BL 5099	3.98	0.65	3.33	0.16	2.31	1.60	0.97	0.14
BL 4984	4.57	0.48	4.09	0.11	2.53	1.48	1.04	0.12
Bhrikuti	3.61	0.77	2.84	0.21	2.19	1.66	0.91	0.15
NL 1437	3.89	0.50	3.40	0.13	2.19	1.39	1.01	0.11
NL 1402	4.46	0.62	3.84	0.14	2.54	1.66	1.00	0.15
Gautam	4.20	0.67	3.53	0.16	2.44	1.68	0.97	0.16
BL 5116	4.29	0.76	3.53	0.18	2.52	1.80	0.95	0.18
NL 1492	4.76	0.65	4.11	0.14	2.71	1.76	1.00	0.17
NL 1488	5.08	0.66	4.42	0.13	2.87	1.83	1.01	0.19
NL 1447	4.43	0.74	3.69	0.17	2.59	1.81	0.96	0.18
NL 1445	4.55	0.66	3.89	0.15	2.60	1.73	0.99	0.17
NL 1506	4.79	0.47	4.32	0.10	2.63	1.50	1.04	0.12
NL 1504	3.53	0.41	3.12	0.12	1.97	1.20	1.02	0.08
NL 1503	4.68	0.32	4.36	0.07	2.50	1.22	1.08	0.08
NL 1501	4.43	0.48	3.95	0.11	2.46	1.46	1.03	0.12
RR 21	3.61	0.37	3.25	0.10	1.99	1.15	1.04	0.07
NL1512	4.47	0.64	3.83	0.14	2.56	1.69	0.99	0.16
NL 1509	4.24	0.38	3.86	0.09	2.31	1.26	1.05	0.09
NL 1508	3.63	0.68	2.95	0.19	2.16	1.57	0.94	0.14

Note: Y_p = Grain yield of genotypes under irrigated condition, Y_s = Grain yield of wheat genotypes under drought condition, TOL = Tolerance index, YSI = Yield stability index, MP = Mean productivity, GMP = Geometric mean productivity, SSI = Stress susceptibility index, STI = Stress tolerance index.



CORRELATION

Table 8. Correlation of Yp, Ys and stress tolerance indices.

	Yp	Ys	TOL	YSI	MP	GMP	SSI	STI
Yp	1							
Ys	0.037	1						
TOL	.955**	-0.262	1					
YSI	-0.367	.911**	-.625**	1				
MP	.957**	0.326	.827**	-0.082	1			
GMP	0.421	.918**	0.134	.675**	.666**	1		
SSI	0.367	-.911**	.625**	-1.000**	0.082	-.675**	1	
STI	0.427	.914**	0.141	.665**	.670**	.998**	-.665**	1

- TOL and MP showed highly significant positive correlation with Yp
- GMP, STI and YSI showed highly significant positive correlation with Ys
- Similar indices were used to identify drought tolerance genotypes by Nyaupane et al., 2023



PRINCIPAL COMPONENT ANALYSIS (PCA)

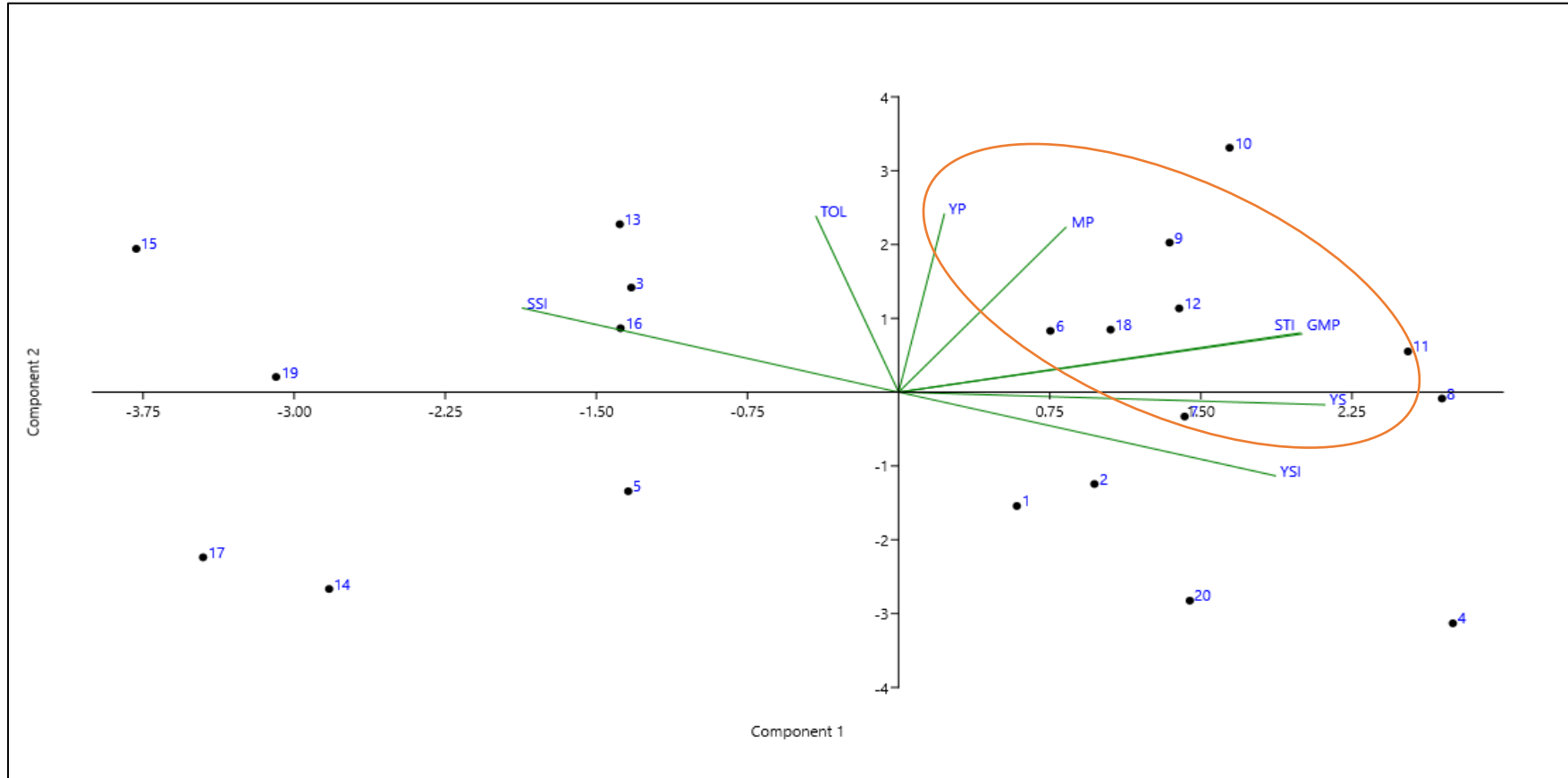


Fig 11. Principal component analysis

- MP, GMP, and STI showed positive correlation with Yp and Ys
- Based on MP, GMP and STI NL 1488 was selected as the drought stress tolerant genotypes



CONCLUSION



- ❑ The wheat yield was reduced by 65.1%, 86.5%, and 45.4% under rainfed, drought, and heat stress, respectively as compared to irrigated condition
- ❑ NL 1488, NL 1506, Bhrikuti, and NL 1488 were high yielding genotypes under irrigated, rainfed, drought, and heat stress, respectively
- ❑ NL 1509, NL 1445, RR21, and Gautam were the most stable genotypes across tested conditions
- ❑ Based on stress tolerance indices NL 1488 was the most drought stress tolerant genotype



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SOME GLIMPSE





Any Questions?

